

WESTERN GEOTECHNICAL CONSULTANTS

Field Inspection Memorandum

To: Mr. David Harmon, P.E., District Engineer
Clearwater Valley Irrigation District

From: R. J. Westbrook, P.E., G.E., Principal Geotechnical Engineer
Western Geotechnical Consultants

Date: March 12, 2019

Project: Cottonwood Creek Pumping Plant (PP-2)
Clearwater Valley Regional Water Conveyance System

WGC Project No.: 2018-GEO-0093

Subject: Foundation Excavation Observation and Bearing Surface Verification

1.0 PURPOSE OF INSPECTION

On March 11 and 12, 2019, WGC personnel visited the PP-2 construction site to observe the foundation excavation and verify bearing surface conditions for the pump bay spread footings and drilled shaft locations. This memorandum documents our observations and recommendations. Mr. K. Nakamura, P.G. (WGC Senior Engineering Geologist) was present on-site for both days of the inspection.

2.0 SITE CONDITIONS AT TIME OF INSPECTION

At the time of our inspection, the general contractor (Boise Valley Constructors, Inc.) had completed the pump bay excavation to a depth of approximately 12 feet below original grade. The excavation was approximately 50 feet long (north-south) by 30 feet wide (east-west), with temporary cut slopes at approximately 1H:1V in the upper clay stratum and 1.5H:1V in the lower sand stratum. Sump pumping was active from two corner sumps, with an estimated combined pumping rate of 40 to 50 gallons per minute. The excavation was stable and no signs of slope instability or excessive seepage were observed.

Weather conditions were clear and dry, with daytime temperatures of approximately 45 to 50 degrees Fahrenheit. No precipitation had occurred in the preceding 5 days. The groundwater level at the excavation floor was being effectively controlled by the sump pumps, with the working surface firm and trafficable.

3.0 SUBSURFACE CONDITIONS OBSERVED

The subsurface conditions exposed in the excavation were consistent with those described in our Geotechnical Investigation Report (Report No. 2018-GEO-0093, October 15, 2018). The following stratigraphic units were observed in the excavation walls:

Layer 1 — Alluvial Fill / Topsoil (0 to approximately 3 feet): Brown, loose, silty sand with gravel and root fragments. This material had been stripped and stockpiled during initial grading operations. Observed thickness was consistent with the 2 to 4 foot range reported in the borings.

Layer 2 — Alluvial Clay (approximately 3 to 11 feet): Olive-brown to brown, firm to stiff, lean clay with occasional thin sand lenses. The clay was exposed in the upper portions of the excavation walls and appeared

uniform in color, consistency, and texture across the entire excavation. Two thin sand lenses (approximately 1 to 2 inches thick) were noted at approximately 7 and 9 feet depth on the east wall. The contact between the clay and underlying sand was gradational over approximately 1 foot, consistent with boring observations. No anomalous soft zones, organic layers, or buried debris were encountered.

Layer 3 — Alluvial Sand and Silt (approximately 11 feet to excavation floor): Gray to tan, medium dense, silty sand. The excavation floor was entirely within this stratum. Groundwater was seeping from the excavation walls at approximately 13.5 to 14 feet below original grade, consistent with the water table depth measured in the 2018 borings (13.5 to 14.2 feet bgs). The sand was medium dense and provided a stable working surface once dewatered.

4.0 BEARING SURFACE VERIFICATION

4.1 Spread Footings

The subgrade for the discharge manifold and motor control center spread footings was examined at four locations along the planned footing alignment. At each location, we performed the following:

- Visual examination of the exposed bearing surface to confirm the material was consistent with the clay stratum (Layer 2) described in the investigation report.
- Hand penetrometer readings on the exposed clay surface. Readings ranged from 2.5 to 3.5 tsf (tons per square foot), indicating firm to stiff consistency consistent with the SPT N-values of 7 to 15 reported for this stratum.
- Verification that the bearing surface was at or below the minimum recommended depth of 4 feet below finished grade.

All four locations were approved for footing placement. The contractor was directed to place the 24-inch structural fill pad immediately after approval to prevent desiccation or disturbance of the bearing surface. No over-excavation beyond the recommended 2-foot swell mitigation depth was required, as the native clay surface was at the proper elevation.

4.2 Drilled Shafts

Six drilled shaft locations were inspected during and after drilling. Shafts were drilled using a truck-mounted Texoma 700 auger rig with a 24-inch diameter auger through the alluvium and a 20-inch rock auger for the bedrock socket. The following observations were made:

Table 1: Drilled Shaft Installation Summary

Shaft No.	Location	Top of Rock (ft)	Socket Depth (ft)	Total Depth (ft)	GW (ft bgs)	Observations
DS-1	NW pump pad	21.5	6.0	27.5	13.8	Clean socket, competent rock
DS-2	NE pump pad	21.0	5.5	26.5	14.0	Clay disc at 24 ft, extended 1 ft
DS-3	Center pump pad	22.0	5.5	27.5	13.5	Clean socket, minor seepage
DS-4	SW support	21.5	5.0	26.5	14.2	Clean socket, competent rock
DS-5	SE support	22.5	5.0	27.5	14.0	Slight weathering in upper 2 ft
DS-6	Manifold pier	22.0	5.5	27.5	13.8	Clean socket, trace clay on joint

Bedrock was encountered at all shaft locations at depths ranging from 21.0 to 22.5 feet bgs, consistent with the 20.5 to 23.0 foot range observed in the 2018 investigation borings. Groundwater was encountered during drilling at depths of 13.5 to 14.2 feet bgs, also consistent with the investigation data.

Shaft DS-2 encountered a clay-filled discontinuity at approximately 24 feet depth within the rock socket. Per our recommendation, the contractor extended the socket an additional 1 foot beyond the minimum 5-foot embedment to ensure adequate bearing below the discontinuity. All other shafts were founded in competent rock at the minimum socket depth.

Each shaft was inspected by WGC personnel prior to concrete placement using a downhole inspection mirror and light. Socket bottoms were clean with less than 0.5 inches of sediment at all locations. Concrete was placed by tremie method in all shafts due to the presence of groundwater. Concrete was specified as 4,500 psi Type II cement mix per the corrosion protection recommendations in the investigation report.

5.0 STRUCTURAL FILL PLACEMENT

The 24-inch structural fill pad beneath the spread footings was placed and compacted on March 12, 2019. The fill material was a well-graded crushed gravel (USCS: GW) obtained from the King Hill Pit, a commercial aggregate source approximately 5 miles south of the site. The material was placed in two 12-inch lifts and compacted with a vibratory plate compactor.

Nuclear density gauge testing was performed on each lift at a frequency of one test per 500 square feet. All test results met or exceeded the specified minimum of 95 percent of the maximum dry density per ASTM D698. Field density results ranged from 96.2 to 98.8 percent relative compaction, with moisture contents within the specified range of optimum minus 2 percent to optimum plus 2 percent.

6.0 CONCLUSIONS AND RECOMMENDATIONS

Based on our observations during the foundation excavation and drilled shaft installation, we conclude the following:

1. The subsurface conditions encountered during construction are consistent with those described in our Geotechnical Investigation Report (Report No. 2018-GEO-0093). No unexpected or adverse conditions were observed.
2. The stratigraphy, layer thicknesses, groundwater depth, and bedrock surface elevation are all within the ranges reported in the investigation. The geologic model developed from the four exploration borings accurately represents the site conditions.
3. All spread footing bearing surfaces were approved as meeting the requirements for an allowable net bearing pressure of 3,000 psf.
4. All drilled shafts were installed to the specified minimum embedment depth in competent bedrock. One shaft (DS-2) required an additional 1 foot of socket depth due to a clay-filled discontinuity, which is consistent with the discontinuity pattern described in the investigation report.
5. Structural fill placement met compaction specifications at all tested locations.
6. The geotechnical recommendations in Report No. 2018-GEO-0093 remain valid for the ongoing construction. No revisions to the foundation design are required based on our construction observations.

We recommend that WGC continue to provide construction monitoring services during the remaining phases of construction, including retaining wall backfill and drainage installation, final grading, and pavement subgrade preparation. Please contact our office if you have questions regarding this memorandum or if conditions change

during the remaining construction.

R. J. Westbrook, P.E., G.E.

Principal Geotechnical Engineer

Idaho PE-9156 / GE-312

cc: K. Nakamura, P.G. (WGC file)

J. Espinoza, Project Manager, Boise Valley Constructors, Inc.